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STATIONARY MEAN FIELD GAMES SYSTEMS DEFINED ON NETWORKS*

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Abstract. We consider a stationary mean field games system constrained on a network. According to the optimal control interpretation of the problem, some transition conditions on the vertices are imposed. We prove separately the well-posedness for each of the two equations composing the system. Finally, we prove existence and uniqueness of the solution of the mean field games system.

Key words. mean field games, networks, transition conditions

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1. Introduction.

The theory of mean field games (MFGs) has been introduced in [21, 24] to describe stochastic differential games with a very large number of agents. Indeed, MFG arise in the study of Nash equilibria for a game with N indistinguishable players, whose individual cost depends on the empirical distribution of all other players, as N tends to $+\infty$. This theory has been successfully applied to economics [2, 10, 20, 17], pedestrian flow [1, 7, 12], and environmental policy [23].

From a mathematical point of view, MFG theory leads to the study of a coupled system of two differential equations: one equation is of Hamilton–Jacobi–Bellman (HJB) type and describes the optimal behavior of the single agent, while the other one is a Fokker–Planck (FP) equation governing the distribution of the overall population. The system can be completed with different boundary conditions (periodic, Dirichlet, Neumann) and initial conditions (initial-terminal condition, planning problem). Existence and uniqueness of strong and weak solutions to the MFG system have been obtained under rather general assumptions on the data of the problems (see [17, 18] and references therein).

In recent times, in view of applications such as data transmission, traffic management, crowd motion, and various problems in biology and neurobiology, there has been an increasing interest in the study of differential models when the dynamics of the problem is constrained on a network. The aim of this note is to study MFG systems defined on a finite network. The optimal control interpretation of the problem will define two differential equations inside each edge (in the usual way) and two *transition* conditions at each vertex. It is worth observing that the structure of the system is similar to the Euclidean case.

Starting with the seminal paper by Lumer [25], a general theory for linear and semilinear differential equations on networks has been developed mainly employing the variational structure of the problem. In this framework the natural transition conditions are, besides the continuity of the solution, the so-called Kirchhoff conditions on the first order derivatives (see [26, 27] and references therein). For a single nonlinear

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equation, existence and uniqueness results are available only for some specific classes of operators such as conservation law [11] and some Hamilton–Jacobi equations [9]. Therefore the first step in our analysis is to establish existence and uniqueness results for each of the two equations composing the MFG system on the network. Because of its control theoretic interpretation (see [14] and section 2) the natural transition condition for the HJB equation is the Kirchhoff condition, while for the FP equation it is natural to require the conservation of the flux at the vertices. In section 2 we discuss the relationship between the transition conditions for the two equations.

After having solved the two equations separately, we tackle our second and main issue: the study of the MFG system on a network. We shall obtain the existence of a solution by a fixed point argument; moreover we shall get uniqueness of the solution adapting a classical argument to this framework and taking advantage of the relation between the transition conditions of the two equations.

As far as we know, this is the first paper to consider general MFG systems (the HJB equation and FP equation as well) on networks. Indeed, in [8] only a particular class of MFG systems on networks was addressed; in that setting, by a suitable change of variables, the HJB and the FP equation are transformed in two heat equations both with Kirchhoff transition conditions coupled via the initial data. Moreover, it is worth observing that the papers [19], [16], and [15] consider MFG systems on graphs (namely, the state variable belongs to a discrete set).

We remark that the results contained here can be generalized in several directions (nonlocal coupling, evolutive problems, boundary conditions, weak solutions, ramified spaces, etc.); moreover the assumptions are far from being optimal. Since this is a first approach to the study of MFG systems on networks, we have tried to keep the presentation as simple as possible in order to avoid technical complications and to concentrate on the network aspects of the problems.

The paper is organized as follows. In the rest of this section, we shall introduce the definition of networks. In section 2 we give a formal derivation of MFG systems on networks and, especially, of the transition conditions. Section 3 is devoted to our main results for the HJB equation, the FP equation, and, mainly, the MFG system. Finally, in Appendix A we collect some technical results.

Networks. The network $\Gamma = (\mathcal{V}, \mathcal{E})$ is a finite collection of points $\mathcal{V} := \{v_i\}_{i\in I}$ in $\mathbb{R}^n$ connected by continuous, non-self-intersecting edges $\mathcal{E} := \{e_j\}_{j\in J}$. Each edge $e_j \in \mathcal{E}$ is parametrized by a smooth function $\pi_j : [0, l_j] \to \mathbb{R}^n$, $l_j > 0$. Given $v_i \in \mathcal{V}$, we denote by $Inc_i := \{j \in J : v_i \in e_j\}$ the set of edges branching out from v_i and by $d_{v_i} := |Inc_i|$ the degree of v_i. A vertex v_i is said a boundary vertex if $d_{v_i} = 1$; otherwise it is said a transition vertex. For simplicity, in this paper we will assume that the set of boundary vertices is empty.

For a function $u : \Gamma \to \mathbb{R}$ we denote by $u_j : [0, l_j] \to \mathbb{R}$ the restriction of u to e_j, i.e., $u(x) = u_j(y)$ for $x \in e_j$, $y = \pi_j^{-1}(x)$, and by $\partial_j u(v_i)$ the oriented derivative of u at v_i along the arc e_j defined by

$$\partial_j u(v_i) = \begin{cases} \lim_{h\to 0^+}(u_j(h) - u_j(0))/h & \text{if } v_i = \pi_j(0); \\ \lim_{h\to 0^+}(u_j(l_j - h) - u_j(l_j))/h & \text{if } v_i = \pi_j(l_j). \end{cases}$$

The integral of a function u on Γ is defined by

$$\int_\Gamma u(x)dx := \sum_{j\in J} \int_0^{l_j} u_j(r)dr.$$

The space $L^p(\Gamma)$, $p \geq 1$, is the set of functions $u : \Gamma \to \mathbb{R}$ such that $u_j \in L^p(0, l_j)$

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for all $j \in J$ and $\|u\|_p := \sum_{j\in J} \|u_j\|_{L^p(0,l_j)} < \infty$. For $p \geq 1$ and for an integer $m > 0$, we define the Sobolev space $W^{m,p}(\Gamma)$ as the space of continuous functions on Γ such that $u_j \in W^{m,p}(0,l_j)$ for all $j \in J$ and $\|u\|_{m,p} := \sum_{j\in J} \|u_j\|_{W^{m,p}(0,l_j)} < \infty$. As usual we set $H^k(\Gamma) := W^{k,2}(\Gamma)$, $k \in \mathbb{N}$. The space $C^k(\Gamma)$, $k \in \mathbb{N}$, consists of all the continuous functions $u : \Gamma \to \mathbb{R}$ such that $u_j \in C^k([0,l_j])$ for $j \in J$ and $\|u\|_{C^k} = \max_{\beta\leq k} \|\partial^\beta u\|_{L^\infty} < \infty$. Observe that no continuity condition at the vertices is prescribed for the derivatives neither for a function $u \in W^{m,p}(\Gamma)$ nor for a function $u \in C^k(\Gamma)$.

Finally, the space $C^{k,\alpha}(\Gamma)$ for $k \in \mathbb{N}$ and $\alpha \in (0,1)$ is the space of functions $u \in C^k(\Gamma)$ such that $\partial^k u_j \in C^{0,\alpha}([0,l_j])$ for any $j \in J$ with the norm

$$\|u\|_{C^{k,\alpha}} := \|u\|_{C^k} + \sup_{j\in J} \sup_{x,y\in[0,l_j],\, x\neq y} [|\partial^k u_j(x) - \partial^k u_j(y)|/|x-y|^\alpha].$$

2. A formal derivation of the MFG system. The MFG system can be deduced from two different points of view (see [24]): either as the characterization of a Pareto equilibrium for dynamic games with a large number of (indistinguishable) players or as the optimality conditions for an optimal control problem whose dynamic is governed by a PDE. We explain the two different points of view for MFG systems on networks showing that they lead to the same transition conditions.

Pareto equilibrium. Consider a population of indistinguishable agents, distributed at time $t = 0$ according to the probability m_0; any agent moves on a network Γ and its dynamics inside the edge e_j is governed by the stochastic differential equation

$$dX_s = -\gamma_s\, ds + \sqrt{2\nu_j}\, dW_s,$$

where γ is the control, $\nu_j > 0$ and W_t is a one-dimensional Brownian motion. When the agent reaches a vertex $v_i \in \mathcal{V}$, it almost surely spends zero time at v_i and enters in one of the incident edges, say, e_j with $j \in Inc_i$, with probability β_{ij}, where

$$\beta_{ij} > 0, \sum_{j\in Inc_i} \beta_{ij} = 1.$$

(See [14] for a rigorous definition of stochastic processes on networks.) The cost criterion is given by

$$\mathbb{E}_x \left[\int_0^T \{L(X_t,\gamma_t) + V[m(X_t)]\} dt + V_0[m(X_T)] \right],$$

where m represents the distribution of the overall population of players. A formal application of the dynamic programming principle gives that the value function u of the previous control problem satisfies

$$(1) \quad \begin{cases} -u_t - \nu_j \partial^2 u + H_j(x,\partial u) = V[m], & (x,t) \in e_j \times (0,T),\ j \in J, \\ \sum_{j\in Inc_i} \alpha_{ij}\nu_j \partial_j u(v_i,t) = 0, & (v_i,t) \in \mathcal{V} \times (0,T), \\ u_j(v_i,t) = u_k(v_i,t), & j,k \in Inc_i,\ (v_i,t) \in \mathcal{V} \times (0,T), \\ u(x,T) = V_0[m(T)], & x \in \Gamma, \end{cases}$$

where $\alpha_{ij} := \beta_{ij}\nu_j^{-1}$ and the Hamiltonian is given on the edge e_j by

$$H_j(x,p) = \sup_\gamma \big[-\gamma \cdot p - L_j(x,\gamma) \big].$$

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Note that the differential equation inside e_j is defined in terms of the coordinate parametrizing the edge. The second equation in (1) is known as the Kirchhoff transition condition, and it is consequence of the assumption on the behavior of X_t at the vertices (see [14]). The third line amounts to the continuity at transition vertices.

In order to derive the equation satisfied by the distribution m of the agents, we follow a duality argument. Consider the linearized Hamilton–Jacobi equation

$$
(2)\quad \begin{cases} -w_t - \nu\partial^2 w + \partial_p H(x,\partial u)\partial w = 0, & (x,t)\in e_j\times(0,T),\ j\in J,\\ \sum_{j\in Inc_i}\nu_j\alpha_{ij}\partial_j w(v_i,t)=0, & (v_i,t)\in\mathcal{V}\times(0,T),\\ w_j(v_i,t)=w_k(v_i,t), & j,k\in Inc_i,\ (v_i,t)\in\mathcal{V}\times(0,T),\\ w(x,T)=0, & x\in\Gamma. \end{cases}
$$

Writing the weak formulation of (2) for a test function m, integrating by parts along each edge, and regrouping the boundary terms corresponding to the same vertex v_i, we get

$$
\begin{aligned}
0 &= \sum_{j\in J}\int_0^T\int_{e_j}\big(-w_t-\nu_j\partial^2 w+\partial_p H_j(x,\partial u)\partial w\big)m\,dxdt\\
&= \sum_{j\in J}\left(\int_{e_j}[wm]_0^T\,dx+\int_0^T\int_{e_j}\big[m_t-\nu_j\partial^2 m-\partial(m\partial_p H_j(x,\partial u))\big]w\,dxdt\right)\\
&\quad -\int_0^T\sum_{v_i\in\mathcal{V}}\Bigg[\sum_{j\in Inc_i}\nu_j m_j(v_i,t)\partial_j w(v_i,t)\\
&\qquad\qquad -\big(\nu_j\partial_j m(v_i,t)+\partial_p H(v_i,\partial u)m_j(v_i,t)\big)w(v_i,t)\Bigg]\,dt.
\end{aligned}
$$

By the previous identity we obtain that m satisfies inside the edges the equation

$$
m_t-\nu\partial^2 m-\partial(m\,\partial_p H(x,\partial u))=0.
$$

Moreover, recalling the transition condition for w, the first one of the three terms computed at the transition vertices vanishes if

$$
(3)\qquad \frac{m_j(v_i,t)}{\alpha_{ij}}=\frac{m_k(v_i,t)}{\alpha_{ik}},\qquad j,k\in Inc_i,\ (v_i,t)\in\mathcal{V}\times(0,T).
$$

The vanishing of the other two terms for each $v_i\in\mathcal{V}$, namely,

$$
(4)\qquad \sum_{j\in Inc_i}\nu_j\partial_j m(v_i,t)+\partial_p H(v_i,\partial u)m_j(v_i,t)=0,
$$

gives the transition condition for m at the vertices $v_i\in\mathcal{V}$. Note that (4) gives the conservation of the total flux of the density m at the vertex v_i (see [11] for a similar

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condition). Summarizing, for $\nu := \{\nu_j\}_{j\in J}$, we get the system

(5)
$$\begin{cases} -u_t - \nu\partial^2 u + H(x, Du) = V[m], & (x,t) \in \Gamma \times (0,T), \\ m_t - \nu\partial^2 m - \partial(m\,\partial_p H(x,\partial u)) = 0, & (x,t) \in \Gamma \times (0,T), \\ \displaystyle\sum_{j\in Inc_i} \alpha_{ij}\nu_j \partial_j u(v_i,t) = 0, & (v_i,t) \in \mathcal{V} \times (0,T), \\ \displaystyle\sum_{j\in Inc_i} \nu_j \partial_j m(v_i,t) + \partial_p H(v_i,\partial u) m_j(v_i,t) = 0, & (v_i,t) \in \mathcal{V} \times (0,T), \\ u_j(v_i,t) = u_k(v_i,t),\ \dfrac{m_j(v_i,t)}{\alpha_{ij}} = \dfrac{m_k(v_i,t)}{\alpha_{ik}}, & j,k \in Inc_i,\ (v_i,t) \in \mathcal{V} \times (0,T), \\ u(x,T) = V_0[m(T)], \quad m(x,0) = m_0(x), & x \in \Gamma, \end{cases}$$

with the normalization condition $\int_\Gamma m(x)dx = 1$. The transition conditions (continuity and either Kirchhoff condition or conservation of total flux) for u and m give d_{v_i} linear conditions for each functions at a $v_i \in \mathcal{V}$, and hence they univocally determine the values $u_j(v_i,t)$ and $m_j(v_i,t)$, $j \in Inc_i$.

Optimal control. We consider the *planning problem* for a MFG system, i.e., we prescribe the initial and the terminal condition for the distribution m (see [3, 24]). Consider the functional

$$\inf_{b,m} \int_0^T \int_\Gamma \{L(x,b)m + W[m]\} dxdt \tag{6}$$

subject to

$$\begin{cases} m_t - \nu\partial^2 m - \partial(bm) = 0, & (x,t) \in \Gamma \times (0,T), \\ \displaystyle\sum_{j\in Inc_i} \nu_j \partial_j m(v_i,t) + b(v_i,t) m_j(v_i,t) = 0, & (v_i,t) \in \mathcal{V} \times (0,T), \\ \dfrac{m_j(v_i,t)}{\alpha_{ij}} = \dfrac{m_k(v_i,t)}{\alpha_{ik}}, & (v_i,t) \in \mathcal{V} \times (0,T), j,k \in Inc_i, \\ m(x,0) = m_0(x), \quad m(x,T) = m_T(x), & x \in \Gamma. \end{cases} \tag{7}$$

The problem of minimizing (6) under the constraints (7) is equivalent to

$$\inf_{b,m} \sup_u \int_0^T \int_\Gamma \{L(x,b)m + W[m] - u(m_t - \nu\partial^2 m - \partial(mb))\} dxdt, \tag{8}$$

where u is the multiplier. We argue as in [3, section 3.3] (see also [18, section 2.5.1]); integrating by parts in (8), taking into account the transition conditions in (7), and minimizing with respect to b, we obtain a minimum problem whose optimality conditions give, at a formal level, a system similar to (5) with an initial-terminal condition for m with $V = W'$.

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Similar considerations in both the approaches can be used for deriving the stationary (*ergodic*) MFG system

$$
(9)\qquad \begin{cases} -\nu\partial^2 u + H(x,\partial u) + \rho = V[m], & x\in\Gamma,\\ \nu\partial^2 m + \partial(m\,\partial_p H(x,\partial u)) = 0, & x\in\Gamma,\\ \displaystyle\sum_{j\in Inc_i} \alpha_{ij}\nu_j\partial_j u(v_i) = 0, & v_i\in\mathcal{V},\\ \displaystyle\sum_{j\in Inc_i} [\nu_j\partial_j m(v_i) + \partial_p H_j(v_i,\partial_j u) m_j(v_i)] = 0, & v_i\in\mathcal{V},\\ u_j(v_i) = u_k(v_i),\ \dfrac{m_j(v_i)}{\alpha_{ij}} = \dfrac{m_k(v_i)}{\alpha_{ik}}, & j,k\in Inc_i,\ v_i\in\mathcal{V},\\ \displaystyle\int_\Gamma u(x)dx = 0, \quad \int_\Gamma m(x)dx = 1, & \end{cases}
$$

where $\rho\in\mathbb{R}$ is also an unknown.

In the rest of the paper we will only consider the stationary system (9). Moreover we will restrict to the case in which all the coefficients in the transition condition for u are equal, i.e.,

$$
(10)\qquad \alpha_{ij} = \alpha_{ik} \qquad \forall i\in I,\ j,k\in Inc_i.
$$

If (10) is not satisfied, the function m should be discontinuous at v_i and this fact clearly involves additional difficulties. In fact it is well known that, in its standard definition, the domain of the Laplace operator on a network is given by the $H^2(\Gamma)$ functions (in particular, continuous) satisfying Kirchhoff condition at the vertices [14, 25]. Moreover, the continuity condition at transition vertices seems to be a crucial ingredient for the comparison principle (see [29, 27]).

3. Main results. This section contains our main results on the solvability of HJB equations, FP equations, and, above all, MFG systems on networks. To this end, we first introduce some assumptions. Consider a Hamiltonian $H:\Gamma\times\mathbb{R}\to\mathbb{R}$, namely, a collection of operators $(H_j)_{j\in J}$ with $H_j:[0,l_j]\times\mathbb{R}\to\mathbb{R}$. For some δ and C positive numbers, we assume

$$
(11)\qquad H_j\in C^2([0,l_j]\times\mathbb{R});
$$

$$
(12)\qquad H_j(x,\cdot) \quad \text{is convex in } p \text{ for each } x\in[0,l_j];
$$

$$
(13)\qquad \delta|p|^2 - C \le H_j(x,p) \le C|p|^2 + C \quad \text{for } (x,p)\in[0,l_j]\times\mathbb{R};
$$

$$
(14)\qquad \nu = \{\nu_j\}_{j\in J}, \quad \nu_j\in\mathbb{R}, \quad \text{with } 0<\nu_0 := \inf_{j\in J}\nu_j.
$$

These assumptions will hold throughout this paper unless it is explicitly assumed in a different way. Let us note that no continuity condition for H is required at the vertices and also that, clearly, the diffusion ν may present discontinuities at these points.

Let us now state our result for MFG systems, whose proof is contained in subsection 3.3; in subsections 3.1 and 3.2 we shall establish our result for HJB equations and, respectively, for FP equations.

THEOREM 1. *Assume* (11)–(14) *and that V is a local C^1 coupling, namely,*

$$
(15)\qquad V[m](x) = V(m(x)) \ \textit{with}\ V\in C^1([0,+\infty)).
$$

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Then, there exists a solution $(u, m, \rho) \in C^2(\Gamma) \times C^2(\Gamma) \times \mathbb{R}$ *to*

$$
(16) \qquad \begin{cases} -\nu\partial^2 u + H(x, \partial u) + \rho = V(m), & x \in \Gamma, \\ \nu\partial^2 m + \partial(m\, \partial_p H(x, \partial u)) = 0, & x \in \Gamma, \\ \displaystyle\sum_{j \in Inc_i} \nu_j \partial_j u(v_i) = 0, & v_i \in \mathcal{V}, \\ \displaystyle\sum_{j \in Inc_i} [\nu_j \partial_j m(v_i) + \partial_p H_j(v_i, \partial_j u) m_j(v_i)] = 0, & v_i \in \mathcal{V}, \\ u_j(v_i) = u_k(v_i),\ m_j(v_i) = m_k(v_i), & j, k \in Inc_i,\ v_i \in \mathcal{V}, \\ \displaystyle\int_\Gamma u(x)dx = 0, \quad \int_\Gamma m(x)dx = 1, \qquad m \geq 0. \end{cases}
$$

Moreover if

$$
(17) \qquad \int_\Gamma (V(m_1) - V(m_2))(m_1 - m_2)dx \leq 0 \Rightarrow\ m_1 = m_2,
$$

then the solution is unique.

Remark 2. The monotonicity condition (17) is quite standard in MFG theory for getting uniqueness. Actually, on one hand it modelizes situations where agents prefer to avoid high density places (as in traffic flow for instance); on the other hand, the uniqueness of the solution to the MFG system may fail when this condition drops (see [24]).

Remark 3. As already pointed out in the introduction, this result can be easily adapted to the case of networks having a boundary by imposing Neumann or Dirichlet boundary condition.

3.1. On the HJB equation. This section is devoted to the ergodic HJB problem: find $(u, \rho) \in C^2(\Gamma) \times \mathbb{R}$ such that

$$
(18) \qquad \begin{cases} -\nu\partial^2 u + H(x, \partial u) + \rho = f(x), & x \in \Gamma, \\ u_j(v_i) = u_k(v_i), \quad \displaystyle\sum_{j \in Inc_i} \nu_j \partial_j u(v_i) = 0, & j, k \in Inc_i, \quad v_i \in \mathcal{V}, \end{cases}
$$

with the normalization condition

$$
(19) \qquad \int_\Gamma u(x)dx = 0.
$$

THEOREM 4. *Assume* (11)–(14) *and* $f \in C^{0,\alpha}(\Gamma)$ *for some* $\alpha \in (0, 1)$. *Then, there exists a unique couple* $(u, \rho) \in C^2(\Gamma) \times \mathbb{R}$ *satisfying* (18)–(19). *Moreover* $u \in C^{2,\alpha}(\Gamma)$, *and there holds*

$$
(20) \qquad \|u\|_{C^{2,\alpha}(\Gamma)} \leq C, \qquad |\rho| \leq \max_\Gamma |H(\cdot, 0) - f(\cdot)|
$$

with C *depending only on* $\|f\|_{C^{0,\alpha}}$ *and the constants in* (13)–(14).

Proof. Proposition 10 ensures that, for any $\lambda \in (0, 1)$, there exists a solution $u_\lambda \in C^{2,\alpha}(\Gamma)$ to

$$
(21) \qquad \begin{cases} -\nu\partial^2 u + H(x, \partial u) + \lambda u = f(x), & x \in \Gamma, \\ u_j(v_i) = u_k(v_i), \quad \sum_{j \in Inc_i} \nu_j \partial_j u(v_i) = 0, & j, k \in Inc_i, \quad v_i \in \mathcal{V}. \end{cases}
$$

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We want to pass to the limit for $\lambda \to 0$ in (21). We first observe that if C_0 is a constant such that $\max_\Gamma |H(\cdot,0)-f(\cdot)| \le C_0$, then the functions $\underline{u}$, $\overline{u}$ defined by $\underline{u}(x) = -C_0/\lambda$, $\overline{u}(x) = C_0/\lambda$ for any $x \in \Gamma$, are, respectively, a sub- and a supersolution of (21). By Proposition 12 we get

$$-C_0 \le \lambda u_\lambda(x) \le C_0 \quad \text{for any } x \in \Gamma. \tag{22}$$

Now, let us introduce the function $w_\lambda := u_\lambda - \min_\Gamma u_\lambda$; it is a $C^{2,\alpha}$ solution to

$$-\nu\partial^2 w_\lambda + H(x,\partial w_\lambda) + \lambda u_\lambda = f(x), \qquad x \in \Gamma, \tag{23}$$

with the same continuity and Kirchhoff transition conditions as in (21). We claim

$$\|\partial w_\lambda\|_{L^2(\Gamma)} \le C_1 \tag{24}$$

for some constant C_1 depending only on $\|f\|_{C^{0,\alpha}}$ and the constants in (13)–(14) (in particular, independent of λ). Indeed, integrating (23) on Γ (i.e., using $\phi = 1$ as test function for w_λ), we get

$$\int_\Gamma H(x,\partial w_\lambda)\,dx + \int_\Gamma (\lambda u_\lambda)\,dx = \int_\Gamma f\,dx.$$

By assumption (13) and estimate (22), we infer

$$\delta \int_\Gamma |\partial w_\lambda|^2\,dx \le \int_\Gamma (C - \lambda u_\lambda + f)\,dx \le (C + C_0 + \|f\|_\infty)|\Gamma|,$$

which amounts to our claim (24). We now claim that

$$\|w_\lambda\|_{C^{2,\alpha}(\Gamma)} \le C_2 \tag{25}$$

for some constant C_2 with the same feature of C_1. To this end, we note that, since w_λ is a classical solution to (23), by (13)–(14) and (22), there holds

$$\nu_0|\partial^2 w_\lambda| \le |H(x,\partial w_\lambda)| + |\lambda u_\lambda| \le C(|\partial w_\lambda|^2 + 1) + C_0$$

and by (24)

$$\|\partial^2 w_\lambda\|_{L^1(\Gamma)} \le C_3 \tag{26}$$

for some constant C_3 sharing the same features of C_1. Taking into account (24) and (26), (possibly increasing C_3) we infer $\|\partial w_\lambda\|_{C^{0,\alpha}(\Gamma)} \le C_3$; using again (23) we accomplish the proof of our claim (25).

Possibly passing to a subsequence, we may assume that, as $\lambda \to 0^+$, the sequence $\{w_\lambda\}_\lambda$ converges to some function $u \in C^{2,\alpha}(\Gamma)$ (observe that u still verifies the continuity and the Kirchhoff conditions) and that $\{\lambda \min_\Gamma u_\lambda\}_\lambda$ converges to some constant ρ. Passing to the limit in (23), we get that the couple (u,ρ) satisfies (18). Possibly adding a constant to u, we also get (19).

To show the uniqueness of ρ, assume that there exist two solutions (u_i,ρ_i), $i = 1,2$, of (18) and let x_0 be a maximum point of $u_1 - u_2$. If $x_0 \in e_j$, we have $\partial_j u_1(x_0) = \partial_j u_2(x_0)$ and $\partial_j^2 u_1(x_0) \le \partial_j^2 u_2(x_0)$. Hence using the equation we conclude that $\rho_2 \le \rho_1$. If $x_0 = v_i$, there holds $\partial_j u_1(x_0) \le \partial_j u_2(x_0)$ for any $j \in Inc_i$. In fact, we have the following: $\partial_j u_1(x_0) = \partial_j u_2(x_0)$ for any $j \in Inc_i$; indeed, assuming by

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contradiction $\partial_k u_1(x_0) < \partial_j u_2(x_0)$ for some $k \in Inc_i$, we get $\sum_{j\in Inc_i} \nu_j \partial_j u_1(x_0) < \sum_{j\in Inc_i} \nu_j \partial_j u_2(x_0)$, which contradicts the Kirchhoff condition. Hence

$$\nu_j \partial_j^2 (u_2 - u_1)(x_0) \geq H(x_0, \partial_j u_2(x_0)) - H(x_0, \partial_j u_1(x_0)) + \rho_2 - \rho_1 = \rho_2 - \rho_1,$$

which, together with $\partial_j u_1(x_0) = \partial_j u_2(x_0)$, gives a contradiction for $\rho_2 > \rho_1$. Hence $\rho_2 \leq \rho_1$ and by symmetry $\rho_2 = \rho_1$.

Having proved the uniqueness of ρ, the uniqueness of a solution to (18)–(19) can be proved as in [9, Corollary 3.1]. Finally, since there exists a unique solution to (18) we conclude that all the sequence $(w_\lambda, \lambda u_\lambda)$ converges to (u, ρ). ∎

3.2. On the FP equation. This section is devoted to the following problem:

$$\begin{cases} \nu \partial^2 m + \partial(b(x)\, m) = f(x), & x \in \Gamma, \\ m_j(v_i) = m_k(v_i), \quad \displaystyle\sum_{j\in Inc_i} [b(v_i) m_j(v_i) + \nu_j \partial_j m(v_i)] = 0, & j,k \in Inc_i,\ v_i \in \mathcal{V}, \end{cases} \tag{27}$$

with the normalization conditions

$$m \geq 0, \qquad \int_\Gamma m(x) dx = 1. \tag{28}$$

Definition 5.

(i) *A strong solution of* (27) *is a function* $m \in C^2(\Gamma)$ *which satisfies* (27) *in pointwise sense.*

(ii) *A weak solution of* (27) *is a function* $m \in H^1(\Gamma)$ *such that*

$$\sum_{j\in J} \int_{e_j} (\nu_j \partial_j m + b(x) m) \partial_j \phi \, dx = \sum_{j\in J} \int_{e_j} f \phi dx \qquad \forall \phi \in H^1(\Gamma). \tag{29}$$

Remark 6. By standard arguments, one can easy check that if $m \in C^2(\Gamma)$ is a weak solution to (27), then it is also a strong solution to (27).

Theorem 7. *Assume* $b \in C^1(\Gamma)$ *and* $f \equiv 0$. *Then, there exists a unique weak solution* m *to* (27)–(28) *and it verifies*

$$\|m\|_{H^1} \leq C, \qquad 0 < m(x) \leq C, \tag{30}$$

where the constant C *depends only on* $\|b\|_\infty$ *and* ν. *Moreover* $m \in C^2(\Gamma)$, *and it is also a strong solution to* (27).

Proof. We shall proceed adapting the arguments of [5, Theorem II.4.3]. By Proposition 13, there exists a unique (up to multiplicative constant) solution to problem (27). So we only have to prove that this family of solutions contains a (unique) function m satisfying (28) which, moreover, will verify (30).

For $\lambda \in (1, +\infty)$ and for any $\phi \in L^\infty(\Gamma)$, we set

$$U_\lambda(t,x) := U(\lambda t, x) \qquad \text{for } (t,x) \in (0,1) \times \Gamma,$$

where U is the solution of the parabolic Cauchy problem (53) with $\psi = \phi$ (see Lemma 14). Observe that U_λ solves

$$\partial_t U_\lambda - \lambda \nu \partial^2 U + \lambda b \partial U = 0 \qquad \text{in } (0,1) \times \Gamma \tag{31}$$

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with the same transition conditions and initial datum of (53). We claim that

$$U_\lambda \text{ are uniformly bounded in } L^2(0,1;H^1(\Gamma)) \cap L^\infty((0,1)\times\Gamma). \tag{32}$$

Indeed, since $\pm\|\phi\|_\infty$ are, respectively, a super- and a subsolution to (31), we get

$$|U_\lambda(t,x)| \le \|\phi\|_\infty \qquad \text{a.e. in } (0,1)\times\Gamma. \tag{33}$$

In other words, we get that the functions U_λ are uniformly bounded in $L^\infty((0,1)\times\Gamma)$ and, in particular, in $L^2((0,1)\times\Gamma)$. On the other hand, using U_λ as test function for problem (31), we get

$$\begin{aligned}
&\int_\Gamma U_\lambda^2(\tau,x)\,dx + \lambda\sum_j \nu_j \iint_{(0,1)\times e_j} (\partial_j U_\lambda)^2\,dxdt \\
&\quad = \int_\Gamma \phi^2(x)\,dx - \lambda \iint_{(0,1)\times\Gamma} b\partial U_\lambda U_\lambda\,dxdt \\
&\quad \le \int_\Gamma \phi^2(x)\,dx + \lambda\|b\|_\infty \iint_{(0,1)\times\Gamma} |\partial U_\lambda||U_\lambda|\,dxdt.
\end{aligned}$$

Applying Cauchy inequality to the last term (recall that Γ has finite measure), by (33), we get

$$\lambda\sum_j \nu_j \iint_{(0,1)\times e_j} (\partial_j U_\lambda)^2\,dxdt \le c(\lambda+1)$$

for some constant c independent of λ. We infer that ∂U_λ are uniformly bounded in $L^2((0,1)\times\Gamma)$; thus our claim (32) is completely proved.

The property (32) yields that there exists a subsequence of $\{U_\lambda\}$ (that we still denote by U_λ) such that

$$U_\lambda \to \xi \quad \text{in } L^\infty((0,1)\times\Gamma) \text{ weak-}* \text{ and in } L^2(0,1;H^1(\Gamma)) \text{ weak as } \lambda\to+\infty.$$

Let us now use $\beta\theta$ as test function for (31) with $\beta\in C_0^\infty((0,1))$ and $\theta\in C^\infty(\Gamma)$ (recall that this means that $\theta\in C^0(\Gamma)$ and $\theta\in C^\infty(e_j)$ for every $j\in J$); we obtain

$$\frac{1}{\lambda}\iint_{[0,1]\times\Gamma} U_\lambda\beta'\theta dx\,dt + \int_{[0,1]}\beta\sum_j\int_{e_j}(\nu_j\partial_j U_\lambda\partial_j\theta + b\partial_j U_\lambda\theta)\,dx\,dt = 0.$$

Passing to the limit as $\lambda\to+\infty$, we get

$$\int_{[0,1]}\beta\sum_j\int_{e_j}(\nu_j\partial_j\xi\partial_j\theta + b\partial_j\xi\theta)\,dx\,dt = 0.$$

By the arbitrariness of β we get that, for a.e. $t\in(0,1)$, there holds

$$\sum_j\int_{e_j}(\nu_j\partial_j\xi\partial_j\theta + b\partial_j\xi\theta)\,dx = 0;$$

namely, ξ is a weak solution to

$$\begin{cases} -\nu\partial^2\xi + b(x)\partial\xi = 0, & x\in\Gamma, \\ \xi_j(v_i) = \xi_k(v_i), \quad \displaystyle\sum_{j\in Inc_i}\nu_j\partial_j\xi(v_i) = 0, & j,k\in Inc_i,\ v_i\in\mathcal{V}. \end{cases}$$

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The maximum principle for this problem (see [27, Theorem 2.1]) ensures that the function ξ is independent of x, namely $\xi = \xi(t)$.

On the other hand, using the function m introduced in Proposition 13 as a test function for (31), we infer

$$\int_\Gamma U_\lambda(\tau, x)m(x)\,dx = \int_\Gamma \phi(x)m(x)\,dx;$$

as $\lambda \to +\infty$, we get

$$\xi(t)\int_\Gamma m(x)\,dx = \int_\Gamma \phi(x)m(x)\,dx.$$

By the arbitrariness of ϕ (recall also that m cannot be identically zero because it belongs to a one-dimensional family), we deduce that $\int m dx$ cannot be zero. Moreover, we also infer that ξ is independent of t, namely, ξ is a constant. By Proposition 13, we can choose m such that $\int m dx = 1$. In conclusion, the last equality reads as

$$\xi = \int_\Gamma \phi(x)m(x)\,dx$$

for every $\phi \in L^\infty(\Gamma)$. (Clearly, ξ depends on ϕ.) By Lemma 14, a standard application of the Kantorovich–Vulikh theorem (see, [26, Theorem 6.3] and the subsequent discussion) ensures that the semigroup associated to the Cauchy problem (53) has a stricly positive integral kernel. Therefore, we may accomplish the proof following the same arguments of [5] and of [4, Lemma 2.3].

Finally, let us prove that m belongs to $C^2(\Gamma)$; fix $j \in J$ and rewrite the equation as

$$\nu\partial_j^2 m = -m\partial_j b - b\partial_j m. \tag{34}$$

Since $b \in C^1(\Gamma)$ and $m \in H^1(\Gamma)$, it follows that $m \in H^2(\Gamma)$, and hence $\partial_j m$ is continuous. Therefore by (34) we conclude that $m \in C^2(\Gamma)$ and it is also a strong solution to (27). ∎

3.3. Proof of Theorem 1. Consider the set

$$\mathcal{K} := \left\{\mu \in C^{0,\alpha}(\Gamma) :\ \mu \geq 0,\ \int_\Gamma \mu dx = 1\right\}$$

and observe that $\mathcal{K}$ is a closed subset of the Banach space $C^{0,\alpha}(\Gamma)$. We define an operator $T : \mathcal{K} \to \mathcal{K}$ according to the scheme

$$\mu \to u \to m \tag{35}$$

as follows. Given $\mu \in \mathcal{K}$, we solve the problem (18)–(19) with $f(x) = V(\mu)(x)$ for the unknowns u and ρ, which are uniquely defined by Theorem 4. Then, for u given, we seek a function m which solves problem (27)–(28) with $b(x) = \partial_p H(x, \partial u)$. By Theorem 7, the function m is univocally defined and we set $m = T(\mu)$. We claim that

$$\text{the map } T \text{ is continuous with compact image.} \tag{36}$$

Let $\mu_n, \mu \in \mathcal{K}$ be such that $\|\mu_n - \mu\|_{C^{0,\alpha}} \to 0$ for $n \to \infty$ and let (u_n, ρ_n), (u, ρ) be the solutions of (18)–(19) corresponding to $f(\cdot) = V(\mu_n(\cdot))$ and, respectively, $f(\cdot) = V(\mu(\cdot))$. By the estimate (20), (possibly passing to a subsequence) u_n converges

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in $C^2(\Gamma)$ to a function $\bar{u}$ and ρ_n converges to some constant $\bar{\rho}$. Since $V(\mu_n) \to V(\mu)$ in $C^{0,\alpha}$ and since the transition and the boundary conditions pass to the limit by the C^2-convergence we get that $\bar{u}$ is a solution of (18)–(19) with $f(x)$ and ρ replaced, respectively, by $V(m)(x)$ and $\bar{\rho}$. By the uniqueness of the solution to (18)–(19) we get that $\bar{u} = u$ and $\rho = \bar{\rho}$; moreover, all the sequence $\{(u_n, \rho_n)\}$ converges to (u, ρ). Let m_n and m be, respectively, the solutions of (27)–(28) with $f = 0$ for $b = \partial_p H(x, \partial u_n)$ and for $b = \partial_p H(x, \partial u)$. By the estimate (30), the functions m_n are equibounded in $H^1(\Gamma)$. Since $\partial_p H(x, \partial u_n)$ uniformly converges to $\partial_p H(x, \partial u)$, passing to the limit in the weak formulation yields that (possibly passing to a subsequence) m_n converges to a solution $\bar{m}$ to (27)–(28) with $b(x) = \partial_p H(x, \partial u)$. Theorem 7 entails that $m = \bar{m}$; hence, the whole sequence $\{m_n\}$ converges to m.

To prove the compactness of the image of T, consider a sequence μ_n such that $\|\mu_n\|_{C^{0,\alpha}} \leq 1$ and let u_n and m_n the functions obtained according to the scheme (35). By (20), $\|u_n\|_{C^{2,\alpha}}$ is uniformly bounded and therefore by (30) also $\|m_n\|_{C^{0,\alpha}}$ is uniformly bounded. As in the proof of Theorem 7, we get an uniform bound on the H^2-norm of m_n for any n. By the compact immersion of $H^2(\Gamma)$ in $H^1(\Gamma)$, we get that the sequence m_n is compact in H^1 and therefore in $C^{0,\alpha}$. Hence, claim (36) is completely proved.

We can therefore conclude by the Leray–Schauder fixed point Theorem that the map T admits a fixed point, i.e., a solution of system (16). Moreover this solution is also smooth by the regularity results in Theorems 4 and 7.

Finally, the uniqueness of the solution to (16) under the assumption (17) follows by a standard argument in MFG theory adapted to the networks (see [24]). We assume that there exists two solutions (u_1, m_1, ρ_1) and (u_2, m_2, ρ_2) of (16). We set $\bar{u} = u_1 - u_2$, $\bar{m} = m_1 - m_2$, $\bar{\rho} = \rho_1 - \rho_2$ and write the equations for $\bar{u}$, $\bar{m}$

$$\begin{cases} -\nu\partial^2\bar{u} + H(x, \partial u_1) - H(x, \partial u_2) + \bar{\rho} - (V(m_1) - V(m_2)) = 0, \\ \nu\partial^2\bar{m} + \partial(m_1\partial_p H(x, \partial u_1) - m_2\partial_p H(x, \partial u_2)) = 0, \\ \sum_{j\in Inc_i} \nu_j\partial_j\bar{u}(v_i) = 0, \\ \sum_{j\in Inc_i} \nu_j\partial_j\bar{m}(v_i) + (m_1\partial_p H(v_i, \partial u_1) - m_2\partial_p H(v_i, \partial u_2)) = 0, \\ \int_\Gamma \bar{m}dx = 0, \quad \int_\Gamma \bar{u}dx = 0. \end{cases}$$

Multiplying the equation for $\bar{m}$ by $\bar{u}$ and integrating over e_j, we get

$$\begin{aligned} &\int_{e_j} \left[-\nu_j\partial_j\bar{u}\partial_j\bar{m} - \big(m_1\partial_p H_j(x, \partial u_1) - m_2\partial_p H_j(x, \partial u_2)\big)\partial_j\bar{u}(x)\right] dx \\ &\quad + \Big[\bar{u}_j(\nu_j\partial_j\bar{m} + m_1\partial_p H_j(x, \partial u_1) - m_2\partial_p H_j(x, \partial u_2))\Big]_0^{l_j} = 0. \end{aligned} \tag{37}$$

Multiplying the equation for $\bar{u}$ by $\bar{m}$ and integrating over e_j, we get

$$\begin{aligned} &\int_{e_j} \nu_j\partial_j\bar{u}\partial_j\bar{m} + \big[H_j(x, \partial_j u_1) - H_j(x, \partial_j u_2) + \bar{\rho} - (V(m_1) - V(m_2))\big]\bar{m}_j dx \\ &\quad + \Big[\nu_j\bar{m}\partial_j\bar{u}\Big]_0^{l_j} = 0. \end{aligned} \tag{38}$$

Adding (37) to (38), summing over $j \in J$, regrouping the terms corresponding to a same vertex v_i, and taking into account the transition and the normalization condi-

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tions for $\bar{u}$ and $\bar{m}$ we get

$$\sum_{j\in J}\int_{e_j}(m_1-m_2)(V(m_1)-V(m_2))dx$$
$$+\sum_{j\in J}\int_{e_j} m_1\big[H_j(x,\partial_j u_2)-H_j(x,\partial_j u_1)-\partial_p H_j(x,\partial_j u_1)\partial_j(u_2-u_1)\big]dx$$
$$+\sum_{j\in J}\int_{e_j} m_2\big[(H_j(x,\partial_j u_1)-H_j(x,\partial_j u_2)-\partial_p H_j(x,\partial_j u_2)\partial_j(u_1-u_2)\big]dx=0.$$

We observe that the first sum amounts to $\int_\Gamma(m_1-m_2)(V(m_1)-V(m_2))dx$. Moreover, by the convexity of H_j and the positivity of m_1 and m_2, the last two sums are nonnegative. Therefore, we have $\int_\Gamma(m_1-m_2)(V(m_1)-V(m_2))\le 0$; assumption (17) ensures $m_1=m_2$. By the uniqueness of the solution to (18) we finally get $u_1=u_2$ and $\rho_1=\rho_2$.

Appendix A. Auxiliary results.

A.1. Auxiliary results for the HJB equation. In this section, we study the following semilinear problem:

$$\begin{cases} -\nu\partial^2 u+H(x,u,\partial u)+\lambda u=0, & x\in\Gamma,\\ u_j(v_i)=u_k(v_i), \quad \displaystyle\sum_{j\in Inc_i}\nu_j\partial_j u(v_i)=0, & j,k\in Inc_i,\ v_i\in\mathcal{V}.\end{cases}\tag{39}$$

As far as we know, this problem has not been tackled before; we shall establish existence, regularity, and uniqueness (via comparison principle).

Definition 8.

(i) *A strong solution of* (39) *is a function* $u\in C^2(\Gamma)$ *which satisfies* (39) *in the pointwise sense.*

(ii) *A weak solution of* (39) *is a function* $u\in H^1(\Gamma)$ *such that*

$$\sum_{j\in J}\int_{e_j}(\nu_j\partial_j u\partial_j\phi+H(x,u,\partial_j u)\phi+\lambda u\phi)dx=0 \qquad \text{for any } \phi\in H^1(\Gamma).\tag{40}$$

Remark 9. One can easily check that, if $u\in C^2(\Gamma)$ is a weak solution of (39), then it is also a strong solution.

Let us now state our existence result.

Proposition 10. *Assume*

$$\begin{aligned}&H_j(\cdot,r,p) \text{ is measurable in } x, \text{ for any } (r,p)\in\mathbb{R}\times\mathbb{R} \text{ and}\\ &H_j(x,\cdot,\cdot) \text{ is continuous in } (r,p), \text{ for a.e. } x\in(0,l_j),\end{aligned}\tag{41}$$

$$|H(x,r,p)|\le C_0+b(|r|)|p|^2 \qquad \text{for a.e. } (x,r,p)\in\Gamma\times\mathbb{R}\times\mathbb{R},\tag{42}$$

$$H(x,r,p) \text{ is not decreasing in } r \text{ for a.e. } (x,r,p)\in\Gamma\times\mathbb{R}\times\mathbb{R},\tag{43}$$

$$\lambda>0,\ \nu_j\in\mathbb{R} \quad \text{with } 0<\nu_0:=\inf_{j\in J}\nu_j,\tag{44}$$

where $C_0>0$, $b:\mathbb{R}\to\mathbb{R}$ *is an increasing function. Then there exists a weak solution to* (39). *Moreover*

$$\|u\|_{H^1}\le C$$

with C *depending on* C_0, λ, *and* ν_0.

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Moreover, if H belongs to $C^{0,\alpha}(\Gamma \times \mathbb{R} \times \mathbb{R})$ for some $\alpha \in (0,1)$, then solution u belongs to $C^{2,\alpha}(\Gamma)$ with

$$\|u\|_{C^{2,\alpha}} \leq C_1(1 + \|u\|_{H^1}), \tag{45}$$

where C_1 is a constant depending only on C_0, b, and ν_0.

LEMMA 11. *Assume* (41), (43), *and, for some $C_H > 0$,*

$$|H(x,r,p)| \leq C_H \quad \text{for } (x,r,p) \in \Gamma \times \mathbb{R} \times \mathbb{R}.$$

Then there exists a weak solution u to (39). *Moreover*

$$\|u\|_\infty \leq \frac{C_H}{\lambda}. \tag{46}$$

Proof. Define a map $T : H^1(\Gamma) \to H^1(\Gamma)$ by taking the weak solution $u = T(v)$ of

$$\begin{cases} -\nu\partial^2 u + \lambda u = -H(x,v,\partial v), & x \in \Gamma, \\ u_j(v_i) = u_k(v_i), \quad \displaystyle\sum_{j \in Inc_i} \nu_j \partial_j u(v_i) = 0, & j,k \in Inc_i,\ v_i \in \mathcal{V}. \end{cases} \tag{47}$$

(Note that the existence of a weak solution to (47) follows by the theory of sesquilinear forms; see, for instance, [27].) Standard estimates imply that T is continuous with a compact image, and hence by Schauder's theorem it admits a fixed point which is a weak solution to (39).

Even though the proof of estimate (46) is standard, for completeness we sketch the argument. Let $G : \mathbb{R} \to \mathbb{R}$ be a smooth function such that $G(t) = 0$ for $t \in (-\infty, 0]$ and G strictly increasing for $t \in (0,\infty)$. Set $K = C_H/\lambda$ and $\phi = G(u - K)$. Then $\phi \in H^1(\Gamma)$ and by taking ϕ as a test function in (40) we get

$$0 = \sum_{j \in J} \int_{e_j} [\nu_j(\partial_j u)^2 G'(u-K) + (H_j(x,u,\partial_j u) + \lambda K)G(u-K) + \lambda(u-K)G(u-K)]dx.$$

Since $H_j(x,u,\partial u) + \lambda K \geq 0$ and $G(u-K) \geq 0$ a.e. on Γ,

$$\sum_{j \in J} \int_{e_j} \lambda(u-K)G(u-K)dx \leq 0,$$

and by $tG(t) \geq 0$ for $t \in \mathbb{R}$, it follows that $(u-K)G(u-K) = 0$ a.e. on Γ, and hence $u \leq K$ a.e. in Γ. ∎

Proof of Proposition 10. The proof of the existence result follows exactly the same argument of the proof of [6, Theorem 2.1], replacing their steps 1 and 2 with Lemma 11. In fact, in the weak formulation of (21), given in Definition 8, the transition condition is transparent and the estimates necessary to prove the result are obtained using the weak formulation (40), which is the same as that of the problem posed in an Euclidean domain.

Consider now $H \in C^{0,\alpha}(\Gamma \times \mathbb{R} \times \mathbb{R})$. We already know that $u \in C^0(\Gamma)$, and we only have to show that $u_j \in C^{2,\alpha}(0,l_j)$ for any $j \in J$. (Recall that u_j is the restriction of u to the edge e_j.) For j fixed, the equation for u (in distributional sense) is

$$-\nu_j \partial_j^2 u = -\lambda u_j - H(x,u_j,\partial_j u) \quad \text{in } (0,l_j). \tag{48}$$

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Since $u_j \in C^0([0,l_j])$ and, by (42), $H(\cdot, u_j(\cdot), \partial_j u(\cdot)) \in L^1((0,l_j))$, by (48) we immediately get $u_j \in W^{2,1}((0,l_j))$ and therefore $\partial_j u \in L^p((0,l_j))$ for any $p \geq 1$. We deduce $H(\cdot, u_j(\cdot), \partial_j u(\cdot)) \in L^p((0,l_j))$ and in particular $u_j \in W^{2,p}((0,l_j))$. Hence $\partial_j u \in C^{0,\alpha}((0,l_j))$ and again by (48) we get the statement. Moreover, the estimate (45) easily follows from (48). ∎

PROPOSITION 12. *Assume* (41) *and* (43)–(44). *Let the functions* $u_1, u_2 \in C^2(\Gamma)$ *satisfy*

$$\text{(49)} \quad \begin{cases} -\nu\partial^2 u_1 + H(x, u_1, \partial u_1) + \lambda u_1 \geq -\nu\partial^2 u_2 + H(x, u_2, \partial u_2) + \lambda u_2, & x \in \Gamma, \\ \sum_{j\in Inc_i} \nu_j \partial_j u_1(v_i) \leq \sum_{j\in Inc_i} \nu_j \partial_j u_2(v_i), & v_i \in \mathcal{V}. \end{cases}$$

Then, $u_1 \geq u_2$ *on* Γ.

Proof. We argue by contradiction assuming $\max_\Gamma(u_2 - u_1) =: \delta > 0$. Let x_0 be a point where $u_2 - u_1$ attains its maximum. The point x_0 either belongs to some edge or it is a vertex. Assume that x_0 belongs to some edge e_j. By their regularity, the functions u_1 and u_2 fulfill

$$u_2(x_0) = u_1(x_0) + \delta, \qquad \partial_j u_2(x_0) = \partial_j u_1(x_0), \qquad \partial_j^2 u_2(x_0) \leq \partial_j^2 u_1(x_0).$$

In particular, we deduce

$$\begin{aligned} &-\nu_j\partial_j^2 u_1(x_0) + H(x_0, u_1(x_0), \partial_j u_1(x_0)) + \lambda u_1(x_0) \\ &\quad \leq -\nu_j\partial_j^2 u_2(x_0) + H(x_0, u_2(x_0), \partial_j u_2(x_0)) + \lambda(u_2(x_0) - \delta) \\ &\quad < -\nu_j\partial_j^2 u_2(x_0) + H(x_0, u_2(x_0), \partial_j u_2(x_0)) + \lambda u_2(x_0), \end{aligned}$$

which contradicts the first relation in (49). Assume that $x_0 = v_i$ for some $v_i \in \mathcal{V}$. Being regular, the functions u_1 and u_2 fulfill $\partial_j u_2(v_i) \leq \partial_j u_1(v_i)$. We claim $\partial_j u_2(v_i) = \partial_j u_1(v_i)$ for each $j \in Inc_i$. In order to prove this equality we proceed by contradiction and assume that $\partial_j u_2(v_i) < \partial_j u_1(v_i)$ for some $j \in Inc_i$. In this case we get $\sum_{j\in Inc_i} \nu_j \partial_j u_2(v_i) < \sum_{j\in Inc_i} \nu_j \partial_j u_1(v_i)$, which contradicts the second inequality in (49); therefore, our claim is proved. Moreover, since $u_1(v_i) = u_2(v_i) - \delta$, we deduce

$$\begin{aligned} H(v_i, u_1(v_i), \partial_j u_1(v_i)) + \lambda u_1(v_i) &= H(v_i, u_2(v_i) - \delta, \partial_j u_2(v_i)) + \lambda(u_2(v_i) - \delta) \\ &< H(v_i, u_2(v_i), \partial_j u_2(v_i)) + \lambda u_2(v_i). \end{aligned}$$

Taking into account the regularity of H and of u_i $(i = 1, 2)$, we infer that in a sufficiently small neighborhood $B_\eta(v_i)$ there holds

$$H(x, u_1(x), \partial u_1(x)) + \lambda u_1(x) < H(x, u_2(x), \partial u_2(x)) + \lambda u_2(x).$$

This inequality and the first relation in (49) entail

$$\begin{aligned} \nu_j\partial_j^2(u_2 - u_1)(v_i) \geq H(v_i, u_2(v_i), \partial_j u_2(v_i)) &- H(x, u_1(v_i), \partial_j u_1(v_i)) \\ &+ \lambda(u_2(v_i) - u_1(v_i)) > 0, \end{aligned}$$

which, together with $\partial_j u_2(v_i) = \partial_j u_1(v_i)$, contradicts that $u_2 - u_1$ attains a maximum in $x_0 = v_i$. ∎

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A.2. Auxiliary results for the FP equation.

PROPOSITION 13. *Assume that $f \in C^0(\Gamma)$ and $b \in C^1(\Gamma)$. The problem* (27) *with $f = 0$ admits a unique (up to multiplicative constant) weak solution. Moreover, for $f \neq 0$, the problem has a solution provided that $\int_\Gamma f dx = 0$.*

Proof. We shall proceed following the technique of [27, Theorem 2.2] and of [5, Theorem II.4.2]. To this end, it is expedient to introduce the following forms on $H^1(\Gamma)$:

$$a(u,v) := \sum_{j\in J} \int_{e_j} (\nu_j \partial_j u \partial_j v + u b_j \partial_j v)\, dx, \qquad (u,v) := \sum_{j\in J} \int_{e_j} u_j v_j dx.$$

We observe that, for $s > 0$ sufficiently large, the form

$$a_s(u,v) := a(u,v) + s(u,v) \tag{50}$$

is coercive on $H^1(\Gamma)$. Actually, the regularity of b entails

$$\begin{aligned} a_s(u,u) &\geq \sum_{j\in J} \int_{e_j} \left[\nu_j (\partial_j u)^2 + s u_j^2 - \|b\|_\infty |u_j||\partial_j u|\right] dx \\ &\geq \sum_{j\in J} \int_{e_j} \left[\frac{\nu_j}{2}(\partial_j u)^2 + \left(s - \frac{\|b\|_\infty^2}{\nu_j}\right) u_j^2\right] dx. \end{aligned}$$

Fix $s > 0$ such that a_s is coercive on $H^1(\Gamma)$. Invoking the Lax–Milgram theorem, we obtain that for every $f \in L^2(\Gamma)$, the problem

$$a_s(u,v) = (f,v) \qquad \forall v \in H^1(\Gamma) \tag{51}$$

has exactly one solution $u =: G_s(f)$. In other words, there exists a map $G_s : L^2(\Gamma) \to H^1(\Gamma)$ such that $G_s(f)$ is the unique solution in $H^1(\Gamma)$ of problem (51). In fact we claim that

$$G_s(f) \in H^2(\Gamma) \qquad \forall f \in L^2(\Gamma).$$

Actually, let us observe that, on each edge e_j, the weak formulation of problem (27) is equivalent to the equality (in distributional sense)

$$\nu_j \partial_j^2 m = f + \partial_j(bm),$$

where the right-hand side is in $L^2(e_j)$. Whence, $m \in H^2(e_j)$ for every $j \in J$ and our claim is completely proved. Let us observe that the weak formulation is equivalent to

$$(I - sG_s)(u) = G_s(f) \qquad \text{in } L^2(\Gamma);$$

indeed, the weak formulation can be written as $a_s(u,v) = (f,v) + s(u,v)$; hence $u = G_s(f) + sG_s(u)$. We observe that, by the Rellich–Kondrachov theorem (see [27]), G_s maps compactly $L^2(\Gamma)$ into itself. By the Fredholm alternative, the existence and the uniqueness of our problem are related to the properties of the operator $(I - sG_s^*)$, where G_s^* is the adjoint operator of G_s.

Let us now calculate G_s^*. To this end, it is expedient to introduce the problem ($s > 0$ is the same as before)

$$\begin{cases} sw - \nu\partial^2 w + b(x)\partial w = h, & x \in \Gamma, \\ w_j(v_i) = w_k(v_i), \quad \sum_{j\in Inc_i} \nu_j \partial_j m(v_i) = 0, & v_i \in \mathcal{V},\ j,k \in Inc_i, \end{cases} \tag{52}$$

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whose weak formulation is

$$\sum_{j\in J}\int_{e_j}(\nu_j\partial_j w\partial_j v + b_j\partial_j wv + swv)\,dx = \sum_{j\in J}\int_{e_j} h_j v dx \qquad \forall v\in H^1(\Gamma).$$

Arguing as before, we infer that there exists a compact map $\tilde{G}_s : L^2(\Gamma)\to H^2(\Gamma)$ such that $\tilde{G}_s(h)$ is the unique weak solution to problem (52).

We claim that $G_s^* = \tilde{G}_s$. In order to prove this claim, it suffices to show that there holds $(G_s(f),h) = (f,\tilde{G}_s(h))$ for every $f,h\in L^2(\Gamma)$. For $m := G_s(f)$ and $z := \tilde{G}_s(h)$, the regularizing effect of G_s and $\tilde{G}_s$ ensures

$$\begin{aligned}(f,\tilde{G}_s(h)) &= \int_\Gamma fz\,dx = \sum_j\int_{e_j}[sm-\nu_j\partial_j^2 m - \partial(bm)]z\,dx\\ &= \sum_j\int_{e_j}[smz+\nu_j\partial_j m\partial_j z + bm\partial_j z]\,dx = \int_\Gamma mh\,dx = \int_\gamma G_s(f)h\,dx,\end{aligned}$$

where the third equality is due to the Kirchhoff condition in (27), while the last two equalities are due, respectively, to the weak definition of (52) and to the definition of m. Hence, we accomplished the proof that $\tilde{G}_s$ is the dual of G_s. Therefore, invoking the Fredholm alternative, we infer that the dimension of the kernel of $(I-sG_s)$ is finite and it coincides with the one of the kernel of $(I-s\tilde{G}_s)$. In order to evaluate the latter, we observe that it is the space of the weak solution to (52) with $h=0$ and $s=0$. For this problem, [27, Theorem 2.1] ensures the maximum principle and in particular that every solution is constant. Therefore the kernel of $(I-sG_s)$ is one-dimensional and the first part of the statement is completely proved.
In conclusion, the Fredholm alternative still guarantees

$$Rg(I-sG_s) = Ker(I-sG_s^*)^\perp;$$

in particular, problem (27) has a solution provided that f is orthogonal to a one-dimensional space. Using $\phi=1$ as test function in (29), we obtain the desired compatibility condition for the solvability of the problem. ∎

LEMMA 14. *For every $\phi\in L^2(\Gamma)$, the parabolic Cauchy problem*

$$\text{(53)}\qquad \begin{cases} \partial_t U - \nu\partial^2 U + b\partial U = 0, & \text{in } (0,+\infty)\times\Gamma,\\ \displaystyle\sum_{j\in Inc_i}\nu_j\partial_j U(t,v_i) = 0, & v_i\in\mathcal{V},\ t\in(0,+\infty),\\ U_j(t,v_i) = U_k(t,v_i), & v_i\in\mathcal{V},\ j,k\in Inc_i,\ t\in(0,+\infty),\\ U(0,x) = \psi(x), & \text{on } \Gamma.\end{cases}$$

admits exactly one weak solution. Moreover, for $\phi\geq 0$, there holds

$$U(t,x)>0 \qquad \forall (t,x)\in(0,+\infty)\times\Gamma.$$

Proof. The existence and uniqueness of the solution are established in [29, Theorem 3.4]. For $\phi\geq 0$, the solution U is strictly positive in $(0,+\infty)\times\Gamma$ because the corresponding semigroup is holomorphic, positive, and irreducible (see [13]). We observe that the positivity of the semigroup is a straightforward consequence of the comparison principle. Moreover, the semigroup is holomorphic because the form a_s introduced in (50) is coercive (as established in the proof of Proposition 13). Finally, the irreducibility of the semigroup can be obtained following the same arguments of [22, Proposition 5.2]. ∎

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Remark 15. Let us recall that estimates for the kernel function for problem (53) can be obtained arguing as in [28].

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